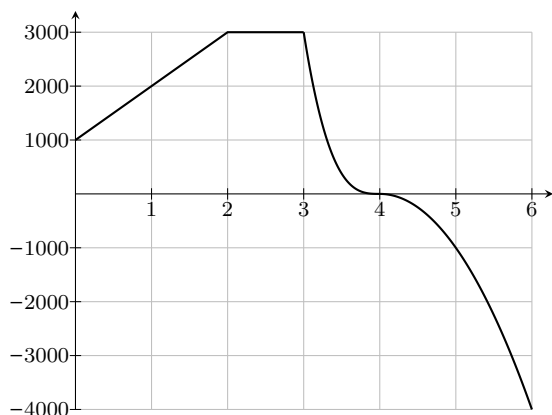


The Fundamental Theorem of Calculus, Part 1

Today's Goals

- ◆ You should be able to visualize accumulation functions of the form $G(x) = \int_a^x f(t) dt$ in terms of signed area and interpret them in terms of net change.
- ◆ You should be able to apply part 1 of the Fundamental Theorem of Calculus, and explain why it makes sense when f is a rate function.
- ◆ You should be able to sketch a graph of $G(x) = \int_a^x f(t) dt$ when given a formula or graph of f .

1. At the Blue Bell ice cream factory, milk is stored in a 40,000 gallon tank. Below is the graph of $f(t)$, the rate (in gallons per hour) at which milk is flowing into and out of the tank t hours after noon one day. At 2 pm, there are 15,000 gallons of milk in the tank.



(a) How much milk is in the tank at 3 pm?

(b) How much milk is in the tank at noon?

(c) Write an expression giving the amount of milk in the tank x hours after noon.

(d) Let $M(x)$ be the amount of milk in the tank x hours after noon. What is $M'(5)$?

(e) Sketch the graph of M .

(f) Let $G(x) = \int_2^x f(t) dt$. In words, what does $G(x)$ represent? Sketch the graph of G on your graph from (e).

(g) Let $H(x) = \int_4^x f(t) dt$. In words, what does $H(x)$ represent? Sketch the graph of H on your graph in (e).

(h) Find $G'(5)$ and $H'(5)$. (Include units.)

(i) **Summarize:** What are the important relationships among f , M , G , and H ?

The Fundamental Theorem of Calculus, Part 1 (FTOC 1)

If f is a continuous function and a is a number, let $G(x) = \int_a^x f(t) dt$. Then $G'(x) = \underline{\hspace{2cm}}$.

2. (a) If $S(x) = \int_3^x \sin\left(\frac{\pi t^2}{2}\right) dt$, what is $S'(4)$?

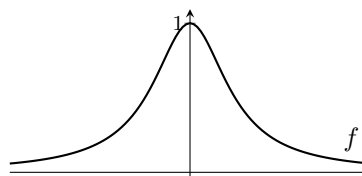
(b) Find $\frac{d}{dx} \left[\int_3^x \sin\left(\frac{\pi t^2}{2}\right) dt \right]$.

(c) Find $\frac{d}{dx} \left(\int_a^b f(x) dx \right)$, where f is a continuous function on $[a, b]$.

3. The graph of a function $f(x)$ is shown below. Let $G(x) = \int_3^x f(t) dt$. Answer the following questions about G , using FTOC 1 as appropriate.

(a) What is $G(3)$?

(b) On what intervals is G increasing? Decreasing? How do you know?



(c) On what intervals is G concave up? Concave down?

(d) Sketch the graph of G .

(e) Let $H(x) = \int_2^x f(t) dt$. Sketch the graph of H on your graph in (d). How does G relate to H ?

4. The rate that water is entering/leaving a tank is given by $f(t) = 60 - 3t$ gallons/minute, where t is measured in minutes past noon. Let $F(x) = \int_0^x f(t) dt$.

(a) Interpret $F(x)$ in words, for $x \geq 0$.

(b) For what values of x is $F(x)$ increasing?

(c) At what time is the water level in the tank the same as it was at noon? At this time, what is the value of $F(x)$?

(d) If the water tank has 30,000 gallons of water at 2 pm, how much does it have at 3 pm? (Be careful with units!)